

ALFIAN NUR USY Aid

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SUMMARY

Computer Science Graduate from STT Terpadu Nurul Fikri (GPA: 3.94) based in Bogor, specializing in scalable software engineering with deep mastery in **OOP**, **SOLID principles**, and **Clean Code** architectures. Experienced in building robust, high-performance web systems and RESTful APIs using **PHP (Laravel)**, **Python (Flask)**, **JavaScript/TypeScript (React, Next.js, Node.js)**, and relational database optimization with **PostgreSQL & MySQL**. Proven track record in academic and faculty-backed research projects, testable backend engineering, and mentorship as a university Teaching Assistant.

TECHNICAL EXPERTISE

- **Core & Architecture:** Object-Oriented Programming (OOP), SOLID Principles, Clean Code, MVC Architecture, Atomic Design Pattern, Component-Driven Development, RESTful API Design, Client-Server Decoupling.
- **Languages:** JavaScript, PHP, Python, TypeScript, SQL (PostgreSQL, MySQL), HTML5, CSS3.
- **Framework & Libraries:** Laravel, Flask, Express.js, Node.js, Next.js, React, Tailwind CSS.
- **Tools & Environment:** Git/GitHub, Postman, Linux CLI, Relational Schema Design.

PROFESSIONAL & ACADEMIC EXPERIENCE

Laboratory Teaching Assistant (Backend, Database, & DSA) | STT Terpadu Nurul Fikri Sep 2024 - Jul 2026

- Facilitated laboratory sessions and mentored students across Backend Engineering (**PHP/Express.js**), RDBMS (**MySQL**), and Data Structures & Algorithms (**Python**).
- Evaluated practicum code repositories and provided 1-on-1 guidance on **OOP principles**, **SQL query optimization**, and complex **algorithmic logic**.

Frontend & Blockchain Researcher | Project: DigiArc Aug 2025 - Sept 2025

- Researched and built decentralized file storage architectures (Web3 Storage) utilizing **Next.js**, **Wagmi**, and **IPFS** data integrity protocols.
- Integrated Solidity smart contracts for cryptographically verified proof-of-storage, adhering to **clean code** conventions and **security standards**.

Fullstack Developer (Academic Research) | Project: ESAO Feb 2025 - Jul 2025

- Engineered ESAO (Essay Analytic Online), an AI-driven automated essay grading platform empowering lecturers to assess essay exams in seconds.
- Architected a decoupled full-stack system utilizing **Next.js** for a responsive analytics dashboard and **Flask (Python)** with **LangChain** for rubric-based, automated student feedback.

Magang dan Studi Independen Bersertifikat (MSIB) | PT Global Investment Institusi (Learning X Academy) Sep 2024 - Dec 2024

| Project : Chicken Yasaka

- Participated in an intensive technical upskilling framework, culminating in the development of "Chicken Yasaka", a full-stack e-commerce platform utilizing **Python (Flask)**, **jQuery (AJAX)**, and **MongoDB**.
- Developed robust business logic including secure user authentication, dynamic cart management, and an admin dashboard.

SOFTWARE ENGINEERING PROJECTS

RoadSense (Smart GIS Navigation) | T3 Stack, Next.js 15, tRPC, Flask, Leaflet, OSRM

- Engineered a web-based Geographic Information System (GIS) application to map road damage through crowdsourced data visualization.
- Implemented complex smart routing algorithms by decoupling the architecture, utilizing **Flask (Shapely)** and **OSRM (Open Source Routing Machine)** for high-performance geospatial calculations.

HRD RESTful API Dashboard | Express.js, Node.js, MySQL, OOP

- Engineered a modular and secure backend architecture for Human Resources management utilizing **Express.js** and **MySQL**.
- Implemented strict MVC patterns with class-based controllers and Data Access Object (DAO) models, ensuring data security through parameterized queries and adherence to RESTful API standards.

L-Movie (Movie Discovery Platform) | React, TailwindCSS, Axios

- Developed a dynamic frontend web application for real-time movie discovery and catalog exploration.
- Implemented asynchronous client-side **REST API** fetching to seamlessly consume and render external database catalogs, ensuring **responsive UI** and **optimal user experience**.

Event Management System | PHP, Laravel (MVC), MySQL

- Developed a comprehensive event registration and management platform applying strict **MVC architectural patterns**.
- Built secure authentication flows, robust role management, and full **CRUD operations** to ensure scalable data handling for event organizers and attendees.

SELECTED CERTIFICATIONS

- **The Complete Web Development Bootcamp** – Angela Yu, Udemy | [Credential Link](#)
 - Skills: Full-Stack Web Development, PostgreSQL, Node.js REST APIs, Relational Database Modeling.
- **Complete JavaScript & Modern Backend Engineering** – Hitesh Choudhary, Udemy | [Credential Link](#)
 - Skills: Core JS Architecture, OOP Design Patterns, Asynchronous Workflows, Clean Code.
- **Certified Competence Full-Stack Web Development** – LearningX Academy | [Credential Link](#)
 - Skills: Full-Stack Application Lifecycle, Clean Architecture, Database Management.

EDUCATION

Bachelor of Computer Science at STT Terpadu Nurul Fikri Jul 2022 - Jul 2026

- **GPA:** 3.94/4.00.
- **Relevant Coursework:** Object-Oriented Programming (OOP), Database Systems, Web Programming, Data Structures & Algorithms.
- **Publication:** Authored and submitted a machine learning prediction model manuscript to the *Multimedia Artificial Intelligent Networking Database Journal* (DOI: <https://doi.org/10.26760/mindjournal.v1i11.115-126>).